



FinTrac — Cloud accounting & operations for SMEs
Full-stack web system: order-to-cash, procure-to-pay, inventory ledger, settlements, statutory-style VAT handling, and financial reporting — aligned with the finflow-smb / finflow-backend codebase.

FYP REPORT STRUCTURE (THESIS CHAPTERS — POSTER SUMMARY)

Introduction
Nepalese SMEs often juggle spreadsheets for stock and informal ledgers for finance. **FinTrac** targets integrated **order-to-cash**, **procure-to-pay**, and **double-entry** posting in one web stack (FinFlow-smb + FinFlow-backend + MySQL), scoped per business **profile**.

Academic questions
How can a single web system improve **consistency** and **auditability** for SMEs versus disconnected tools? Sub-questions: Can document lifecycles (draft → approve → post) align **Inventory** with **GL**? Can **reconciliation** be subledger to control accounts?

Literature review
Academic and industry sources converge on **cloud accounting adoption** (simplicity, cost, trust, regulatory readiness), **Inventory-GL integration** for accurate COGS and planning, **receivables** and liquidity risk, **security perception** as an adoption barrier, and **usability / CSV migration** from spreadsheets. FinTrac is positioned as an integrated O2C/P2P + inventory + settlements prototype that speaks directly to those gaps.

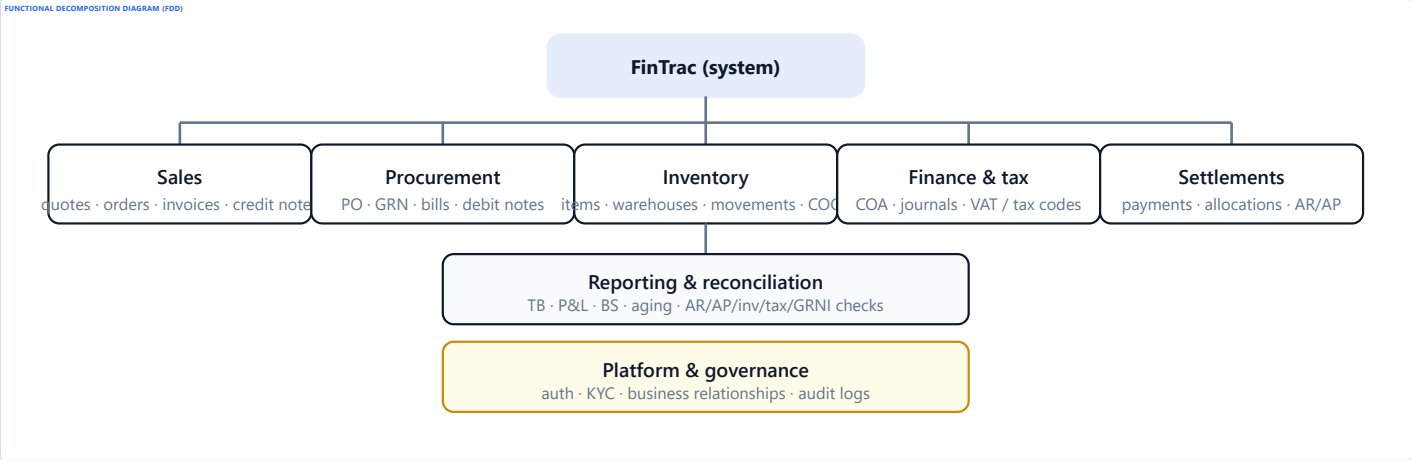
Functional decomposition diagram (FDD)
Top-level capabilities map to src/App, src/routes and FinFlow-backend/services/*. See the FDD figure directly below this block.

Gantt chart
18-week master schedule: blue = implementation; amber = thesis writing for Introduction, Academic Questions, Literature Review, this Gantt chapter, FDD, Artifact description, Conclusion, and Future Scope. See the combined chart in **Testing approach** below.

Artifact developed
Deliverable working **FinTrac** application — React (TypeScript SPA, Express 5 API, JWT auth), KYC flow, automated tests (FinFlow-backend/tests), and MySQL schemas for documents, journals, stock movements, settlements, and reports.

Conclusion
FinTrac demonstrates a **coherent SMB workflow** from operational documents to **posted journals** and **management reports**, without a full multinational ERP — suitable as an FYP-scale **proof of concept**.

Future scope
Multi-user roles per company, **maker-checker** enforcement, CSV import/report for items and movements, **bank feeds**, **payroll**, mobile offline, and hardened **tenant isolation** beyond profile-scoped DBs.



OVERVIEW

FinTrac (branded in the React UI; repository **finflow-smb**) is a browser-based business suite for small and medium enterprises. It replaces ad-hoc spreadsheets for sales and purchasing cycles with **document-driven workflows**, a **MySQL** data store, and **double-entry journals** generated when operational documents post. Tenancy is rooted in authenticated **profiles** (each business user / company scope), with optional migration toward a separate companies entity where present.

- **Frontend:** React (TypeScript), Vite, React Router, TanStack Query, shadow-style UI components.
- **Backend:** Node.js, Express 5, mysql2 (promise pool + legacy callback connection), JWT sessions, bcrypt password hashing.
- **Compliance-style onboarding:** KYC document upload, kyc_status gate, separate **admin** portal for review.
- **Signup security:** Email OTP via pending_signups when SMTP/Gmail is configured (bypassed for direct DB seed accounts in development).

AIMS & ACADEMIC QUESTION

Aims

- Unify **commercial documents** (quotes through invoices, POs through bills) with **stock** and **GL impact**.
- Give SMEs **AR/AP aging**, **trial balance / P&L / balance sheet**, and **reconciliation** views over subledgers.
- Support **inter-business relationships** (buyer/seller invites) on a shared platform.

Academic question
How can a single integrated web system improve **data consistency** and **auditability** for Nepalese-style SMEs compared to disconnected inventory spreadsheets and manual ledger books — while remaining usable without an enterprise ERP team?

OBJECTIVES (MAPPED TO IMPLEMENTED MODULES)

- **Sales:** Quotes → Orders → **Invoices** → **Credit notes**; optional **approval** before posting.
- **Procurement:** **Purchase orders** → **Goods receipts** (stock / GRNI) → **Purchase bills** → **Debit notes**.
- **Inventory:** Items, warehouses, stock movements; weighted-average costing; COGS on posted stock sales.
- **Finance:** chart_of_accounts, journals, entries, tax_codes (e.g. VAT), **settlements** / payment allocation.
- **Reporting:** Trial balance, P&L, balance sheet, AR/AP aging, stock summary, customer/vendor statements, reconciliation bundle (AR, AP, inventory, tax advances, GRNI).

CORE USER FLOWS (ACCURATE TO SERVICES)

Order-to-cash (sales)

Sales quote (draft) + Sales order (e.g. convert from quote) + Sales invoice (draft)
+ submit for approval + approve + POST
+ Journal: Revenue, AR, output VAT (via tax service)
+ stock: sale_issue movements + COGS / Inventory lines when lines reference stocked items

Procure-to-pay

Purchase order + Goods receipt (posted) + stock / GRNI per line accounting
+ Purchase bill (match receipt where applicable) + POST
+ AP, purchase/revenue, input VAT; avoids duplicate receipt when billed free GR

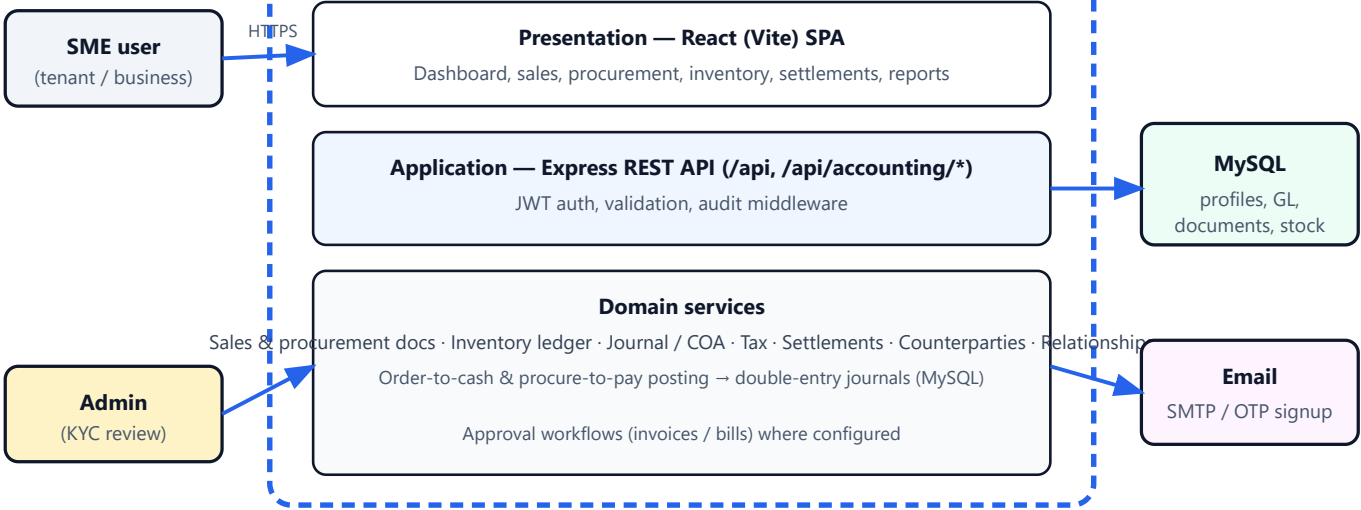
Settlements

Payments (incoming/outgoing) + allocations to invoices/bills; customer/vendor advances when overpaid.

SYSTEM CONTEXT DIAGRAM (LEVEL 0)

FinTrac sits between **human actors** and **external systems**. The boundary matches FinFlow-smb (React) + FinFlow-backend (Express) + MySQL.

FinTrac — System



UI routes: /sales/*, /procurement/*, /inventory, /settlements, /relationships, /reports, /settings (src/App.tsx)

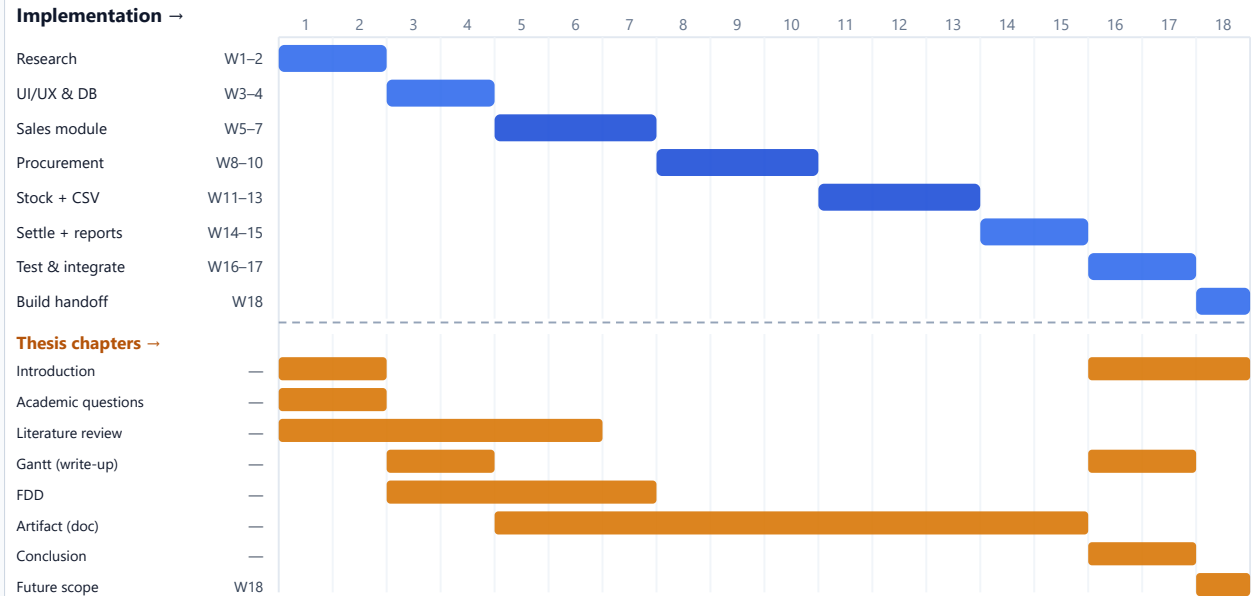
TESTING APPROACH

- Automated: non-text in E2EFlow-backend (Node.js built-in test runner) — integration-style tests for order-to-cash, procurement, inventory posting, settlements, counterparty resolution, reconciliations, and approval gating.
- Manual / API: Postman or browser network tab against running backend; verify posting, stock, and report endpoints.
- UI: Manual walkthrough of document statuses (draft → submitted → approved → posted) and KYC/Admin paths.

PROJECT TIMELINE (18 WEEKS) — IMPLEMENTATION + THESIS CHAPTERS

Blue = software work; amber = written report sections (Introduction, Academic Questions, Literature Review, Gantt write-up, FDD, Artifact description, Conclusion, Future Scope) drafted and revised across the same 18 weeks.

Implementation: Thesis / report chapter



Implementation: W1-2 research · W3-4 UI/UX & DB · W5-7 sales · W8-10 procurement · W11-13 stock (+ CSV) · W14-15 settlements & financial reports · W16-17 testing & integration · W18 build handoff / buffer
Thesis (amber bars): Introduction — draft W1-2, finalize W17-18; Academic questions — W17-18; Literature review — W1-2; Gantt chapter — plan W5-6, update for final report W16-17; FDD — W5-7 with design & early modules; Artifact developed (written description of software) — W5-15 parallel to coding; Conclusion — W17-18; Future scope — W18

LITERATURE REVIEW — THEMATIC SYNTHESIS & FINTRAC POSITIONING

The written review synthesizes SME-focused work on **cloud accounting adoption** (e.g. usefulness, cost, vendor trust, regulatory awareness — Alhamaila et al. 2013; Majidi 2024; OECD 2022), **inventory linked to the general ledger** (Romney & Steinbart 2017; Liang & Wu 2019), **receivables and liquidity** (Atsadius 2023; Rawashdeh 2023), **security and trust** (Nagahawatta 2021; Chen & Zhu 2020), and **onboarding from spreadsheets** (Square 2022; Ch7 2022; Papa et al. 2018). FinTrac implements an **operational + financial** web stack that responds to those themes; the table below maps literature themes to this artifact (full Harvard references in the thesis chapter).

Topic	Comparison / insight
Cloud adoption & barriers	Literature links uptake to perceived benefits but also to trust, security perception, and digital maturity . FinTrac is a low-infrastructure, browser-based prototype aimed at reducing adoption friction for SME-style workflows.
Inventory-accounting integration	Studies to automated stock-ledger flows to accuracy and purchasing efficiency. FinTrac posts inventory movements from sales, credit notes, goods receipts, and bills into a coherent ledger alongside double-entry journals.
Receivables & cash visibility	Industry and survey work highlight late payment and weak monitoring as liquidity risks; cloud accounting can improve recovery metrics. FinTrac supports invoice / settlement flow, aging-style reporting, and financial dashboards grounded in posted data.
Security, usability, CSV	Security concerns suppress adoption unless controls are visible. CSV / structured import and simple UX ease migration off spreadsheets. The stack uses JWT auth , tenant-scoped APIs, and CSV-capable paths aligned with the full review's recommendations.

EVALUATION / REFLECTION

Building FinTrac required reconciling **legacy tables** (inventory, clients, vendors) with **modern** items/stock movements and **canonical counterparties**. The most demanding work was transactional posting (rollback on inventory or account-resolution failure) and keeping **document sequences** and **journal entry numbers** unique per tenant. The project strengthened skills in full-stack design, schema migration under live routes, and test-driven regression for accounting invariants.

CONCLUSION

FinTrac delivers a **coherent SME workflow** from commercial documents through to **posted journals, tax transactions, and management reports**, with inventory tied to sales and procurement events. It demonstrates how a focused web architecture (React + Express + MySQL) can improve trackability for smaller businesses without adopting a full multinational ERP.

FUTURE SCOPE

- Multi-user roles per company and stricter maker-checker on approvals.
- CSV / Excel import-export for items, stock movements, and opening balances.
- Bank statement import, payroll, and richer mobile / offline use.
- Stronger tenant isolation and optional marketplace integrations.